# Calcurion Maths Worksheets

A-Level Mathematics

**Topic:** Mechanics – Connected Particles

Medium & Hard • 25 Questions

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## Section A – Medium (Q1–15)

**1.** Two particles of masses 3 kg and 5 kg are connected by a light inextensible string over a smooth pulley. Find the acceleration and the tension. **[5]**

**2.** Two particles of masses 2 kg and 3 kg are connected by a light string over a smooth pulley. Find the acceleration and tension. **[5]**

**3.** Particles of masses 4 kg and 6 kg connected over a smooth pulley. Find acceleration and tension. **[5]**

**4.** A particle of mass 2 kg on a smooth horizontal table is connected by a light string over a smooth pulley to a particle of mass 3 kg hanging freely. Find the acceleration and tension. **[5]**

**5.** A 5 kg particle on a smooth table is connected to a 2 kg hanging particle. Find acceleration and tension. **[5]**

**6.** Two particles $m$ and $2m$ connected over a smooth pulley. Find acceleration in terms of $g$. **[4]**

**7.** Particles of masses 3 kg and 7 kg connected over a smooth pulley. The system is released from rest. Find the speed of each particle after 2 s. **[5]**

**8.** A 4 kg particle on a smooth table is connected to a 1 kg hanging particle. Find the tension and the force on the pulley. **[5]**

**9.** Two particles of equal mass $m$ are connected over a smooth pulley. What is the acceleration? **[2]**

**10.** A particle of mass 3 kg on a smooth plane inclined at 30° is connected over a smooth pulley to a hanging particle of mass 2 kg. Find the acceleration. **[6]**

**11.** Particles 2 kg and 5 kg connected over a smooth pulley. Find how far the 5 kg mass descends in 1.5 s from rest. **[5]**

**12.** A 6 kg mass on a smooth table is connected to a 4 kg hanging mass. Find the tension. **[4]**

**13.** Two particles of masses 1 kg and 4 kg connected over a smooth pulley. Find the acceleration and the tension. **[5]**

**14.** A particle of mass 3 kg is connected by a light string to a particle of mass 5 kg. The string passes over a smooth pulley. The 5 kg mass is 2 m above the ground. Find the speed of the 5 kg mass when it hits the ground. **[5]**

**15.** Two particles of masses $m$ and $3m$ connected over a smooth pulley. Find the tension in terms of $m$ and $g$. **[4]**

## Section B – Hard (Q16–25)

**16.** Particles of masses 2 kg and 3 kg are connected over a smooth pulley. The 3 kg mass is on a smooth plane inclined at 30°, the 2 kg hangs freely. Find the acceleration. **[6]**

**17.** A 4 kg particle on a rough horizontal table ($\mu = 0.2$) is connected to a 3 kg hanging particle. Find the acceleration. **[6]**

**18.** Two particles of masses 5 kg and 3 kg are connected over a smooth pulley. After the system has been moving for 2 s, the string breaks. Find the speed of each particle at that instant and describe subsequent motion of the 3 kg mass. **[7]**

**19.** A particle of mass 2 kg on a smooth plane inclined at $40°$ is connected over a pulley to a hanging particle of mass 3 kg. Find which way the system moves and the acceleration. **[6]**

**20.** Particles $A$ (2 kg) and $B$ (5 kg) are connected by a light string over a smooth pulley. $A$ is held on the ground with the string taut and $B$ is 1.5 m above the ground. $A$ is released. Find the speed of $B$ when it hits the ground and the time taken. **[6]**

**21.** Three particles of masses $m$, $2m$ and $3m$ are connected by light strings over two smooth pulleys in a line. Find the accelerations. **[7]**

**22.** A 5 kg mass on a smooth table is connected to two hanging masses of 2 kg and 3 kg on either side. Find the acceleration of the system. **[6]**

**23.** Particles of masses 4 kg and 6 kg connected over a smooth pulley. The string is cut when the speeds are $3\,\mathrm{m\,s^{-1}}$. Find the additional distance each travels before coming to instantaneous rest (the heavier one continues down). **[6]**

**24.** A particle of mass 3 kg rests on a smooth plane inclined at $25°$ and is connected over a pulley to a hanging mass of 2 kg. Find the force on the pulley when the system is released. **[6]**

**25.** Show that for two particles of masses $m_1 > m_2$ connected over a smooth pulley, the acceleration is $\dfrac{(m_1 - m_2)g}{m_1 + m_2}$ and the tension is $\dfrac{2m_1 m_2 g}{m_1 + m_2}$. **[5]**

*End of Worksheet* • *Show all working clearly. Take* $g = 9.8\,\mathrm{m\,s^{-2}}$ *unless otherwise stated.*